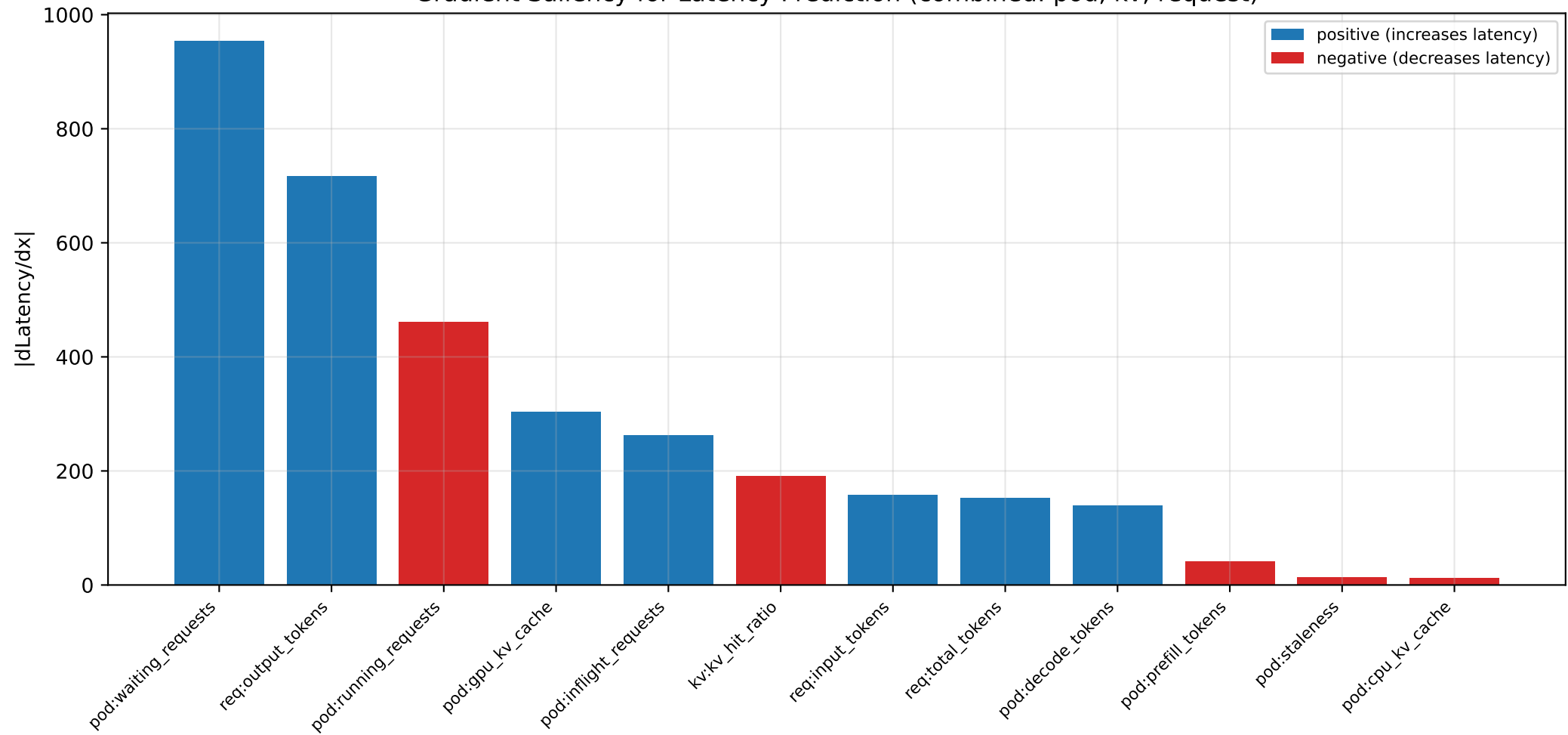


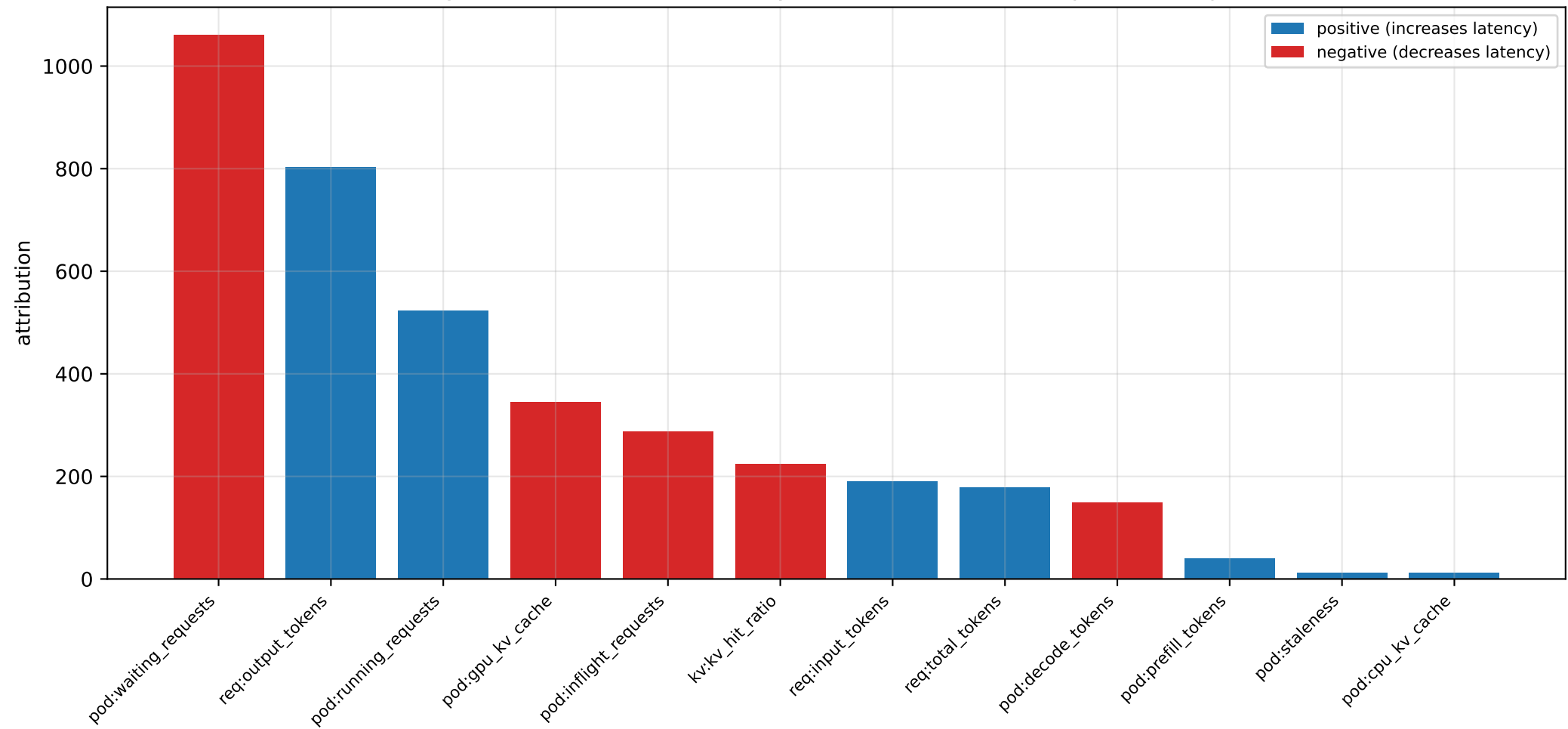
Latency Predictor Explainability Report

final_model-latency_predictor_ttft

Gradient Saliency for Latency Prediction (combined: pod, kv, request)



Integrated Gradients for Latency Prediction (combined: pod, kv, request)



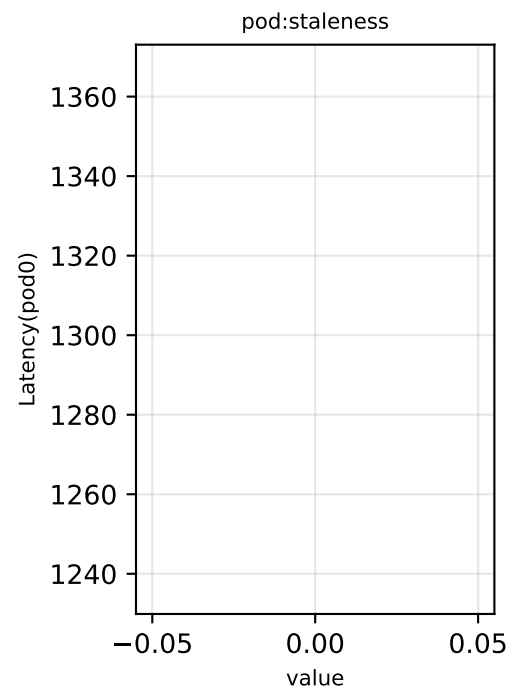
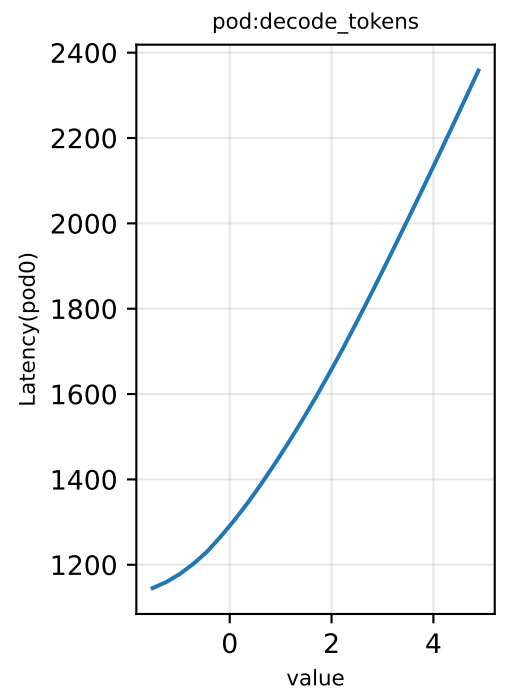
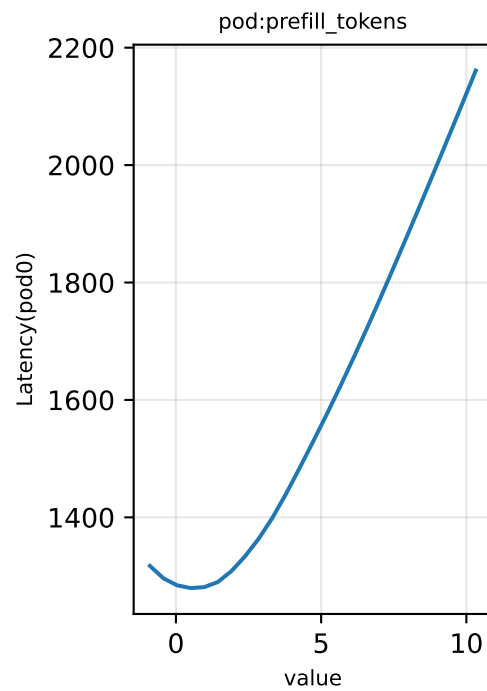
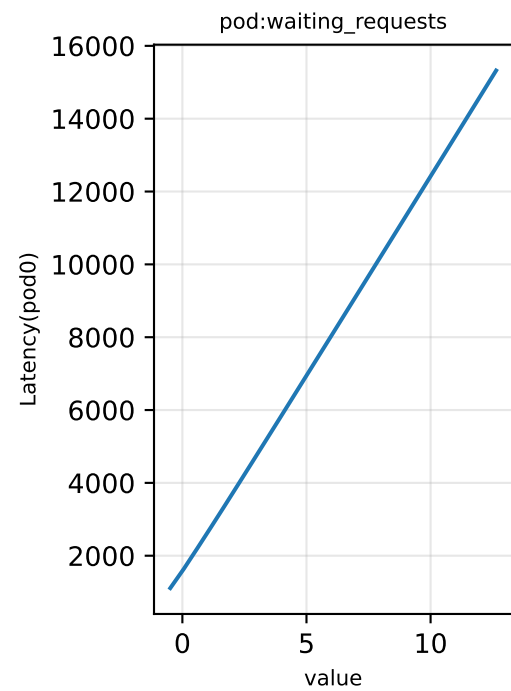
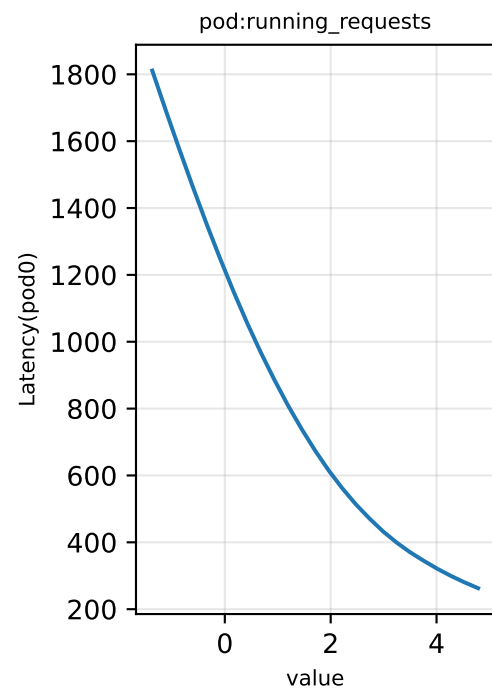
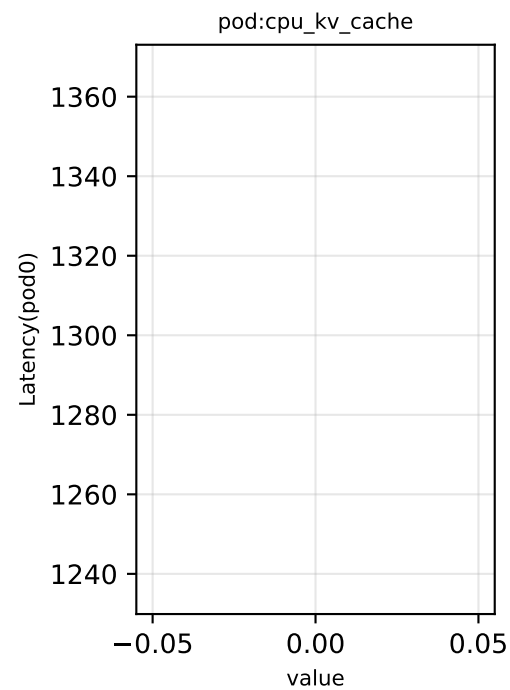
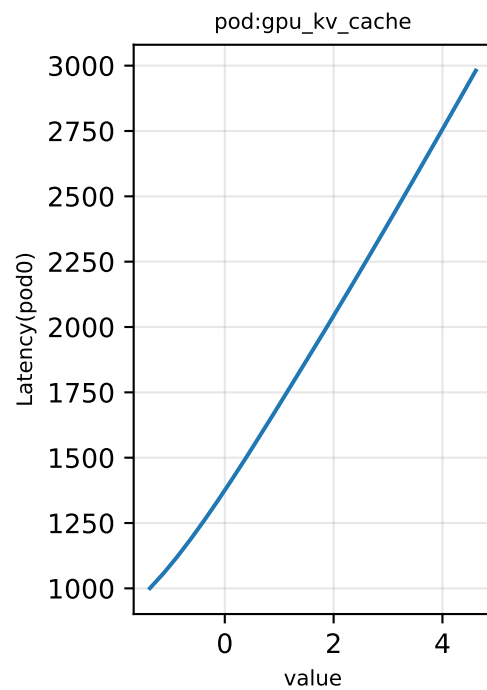
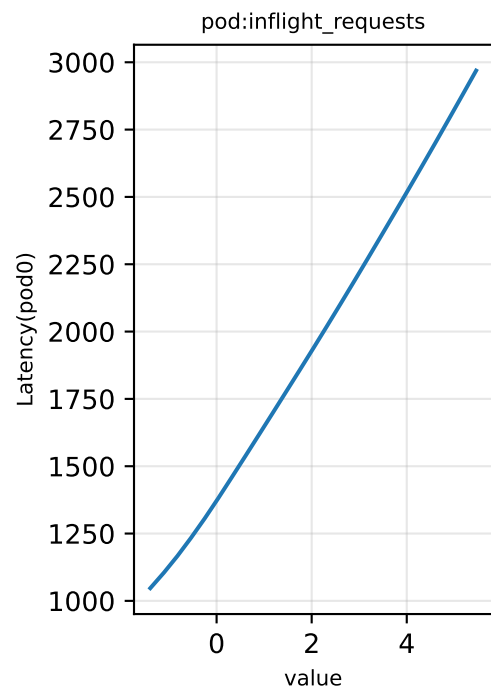
Permutation Symmetry Test

match_rate: 1.0

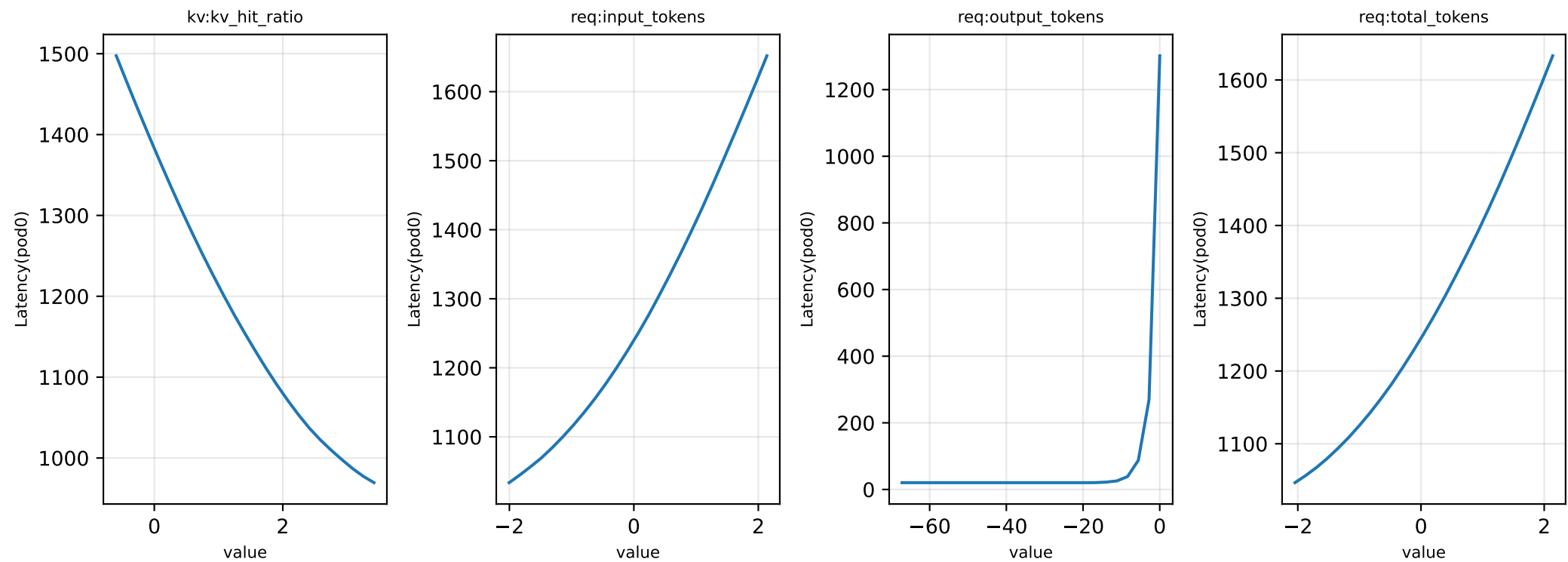
mean_l1_diff: 5.63848601814243e-05

Higher match_rate and lower mean_l1_diff indicate invariance to pod ordering.

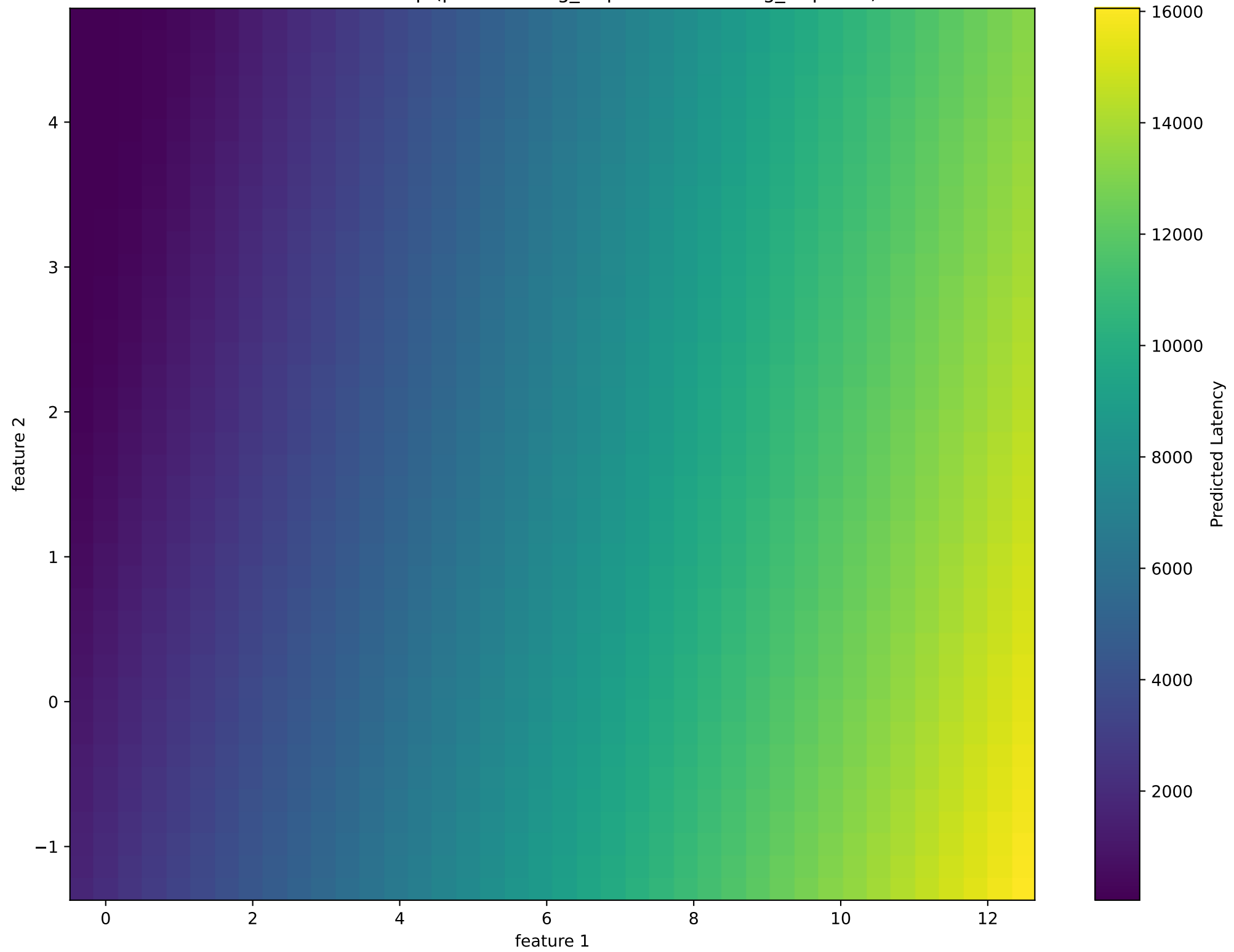
PDPs (all groups), target pod=0



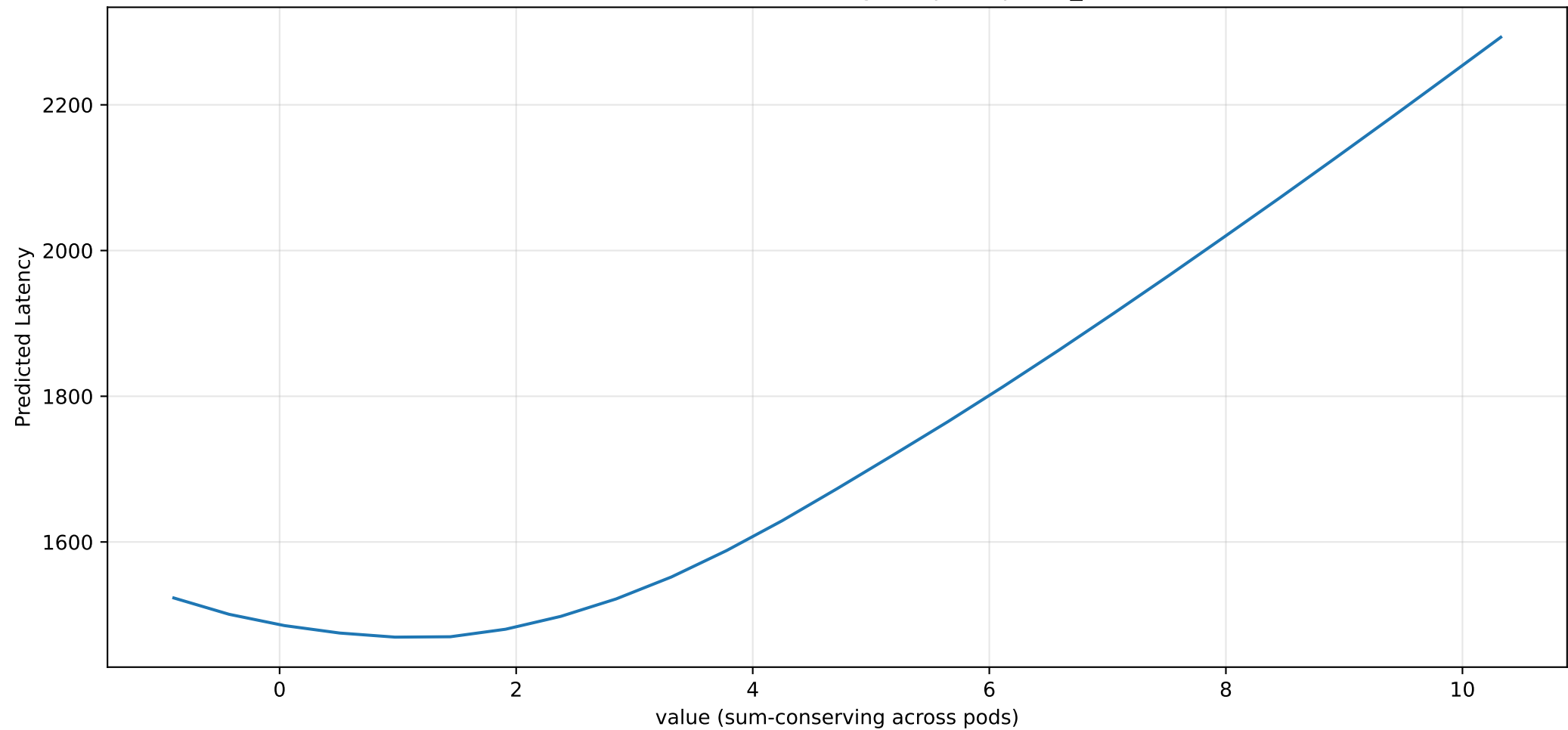
PDPs (all groups), target pod=0



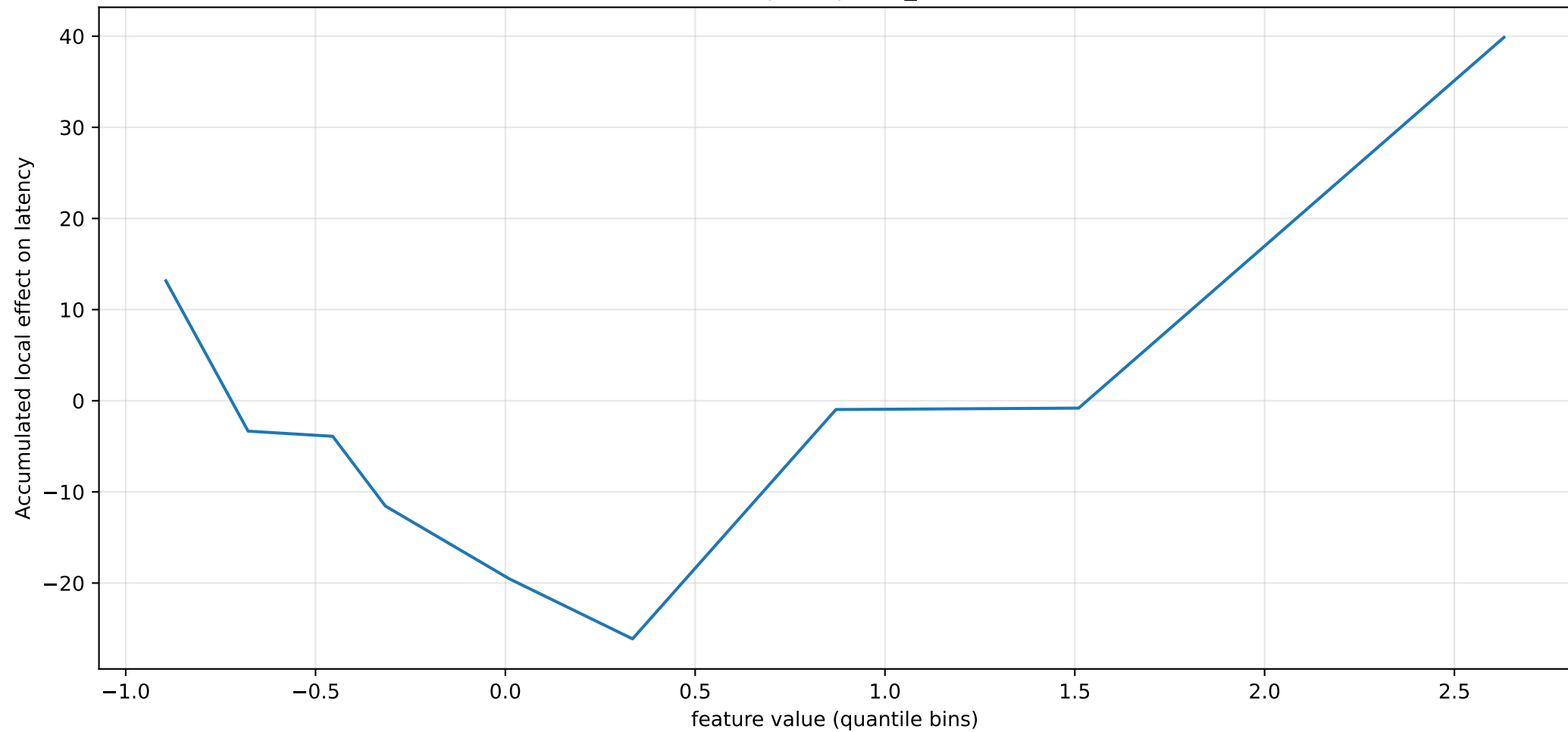
Interaction heatmap (pod0 waiting_requests vs running_requests)



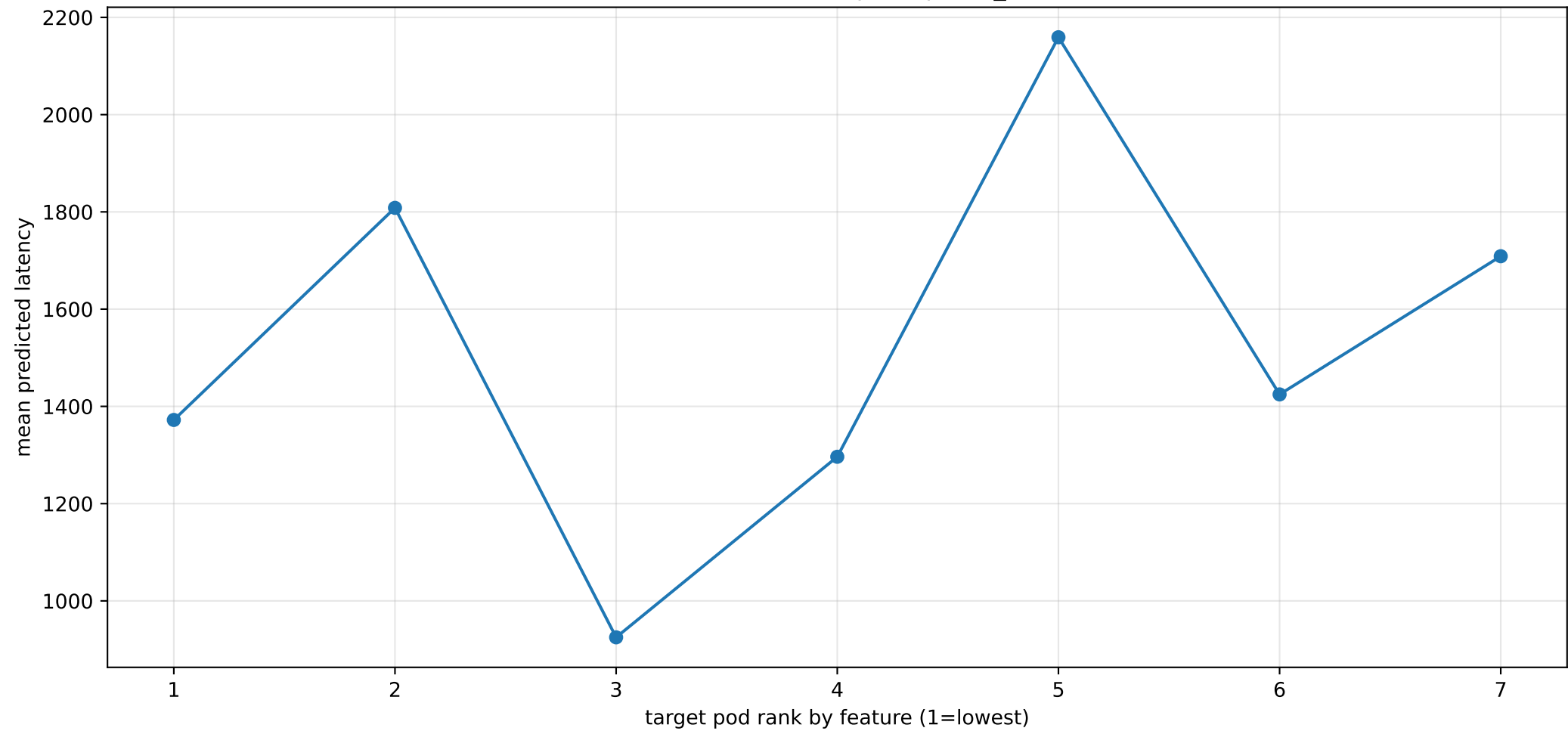
Relative PDP (sum-conserving) for pod0:prefill_tokens



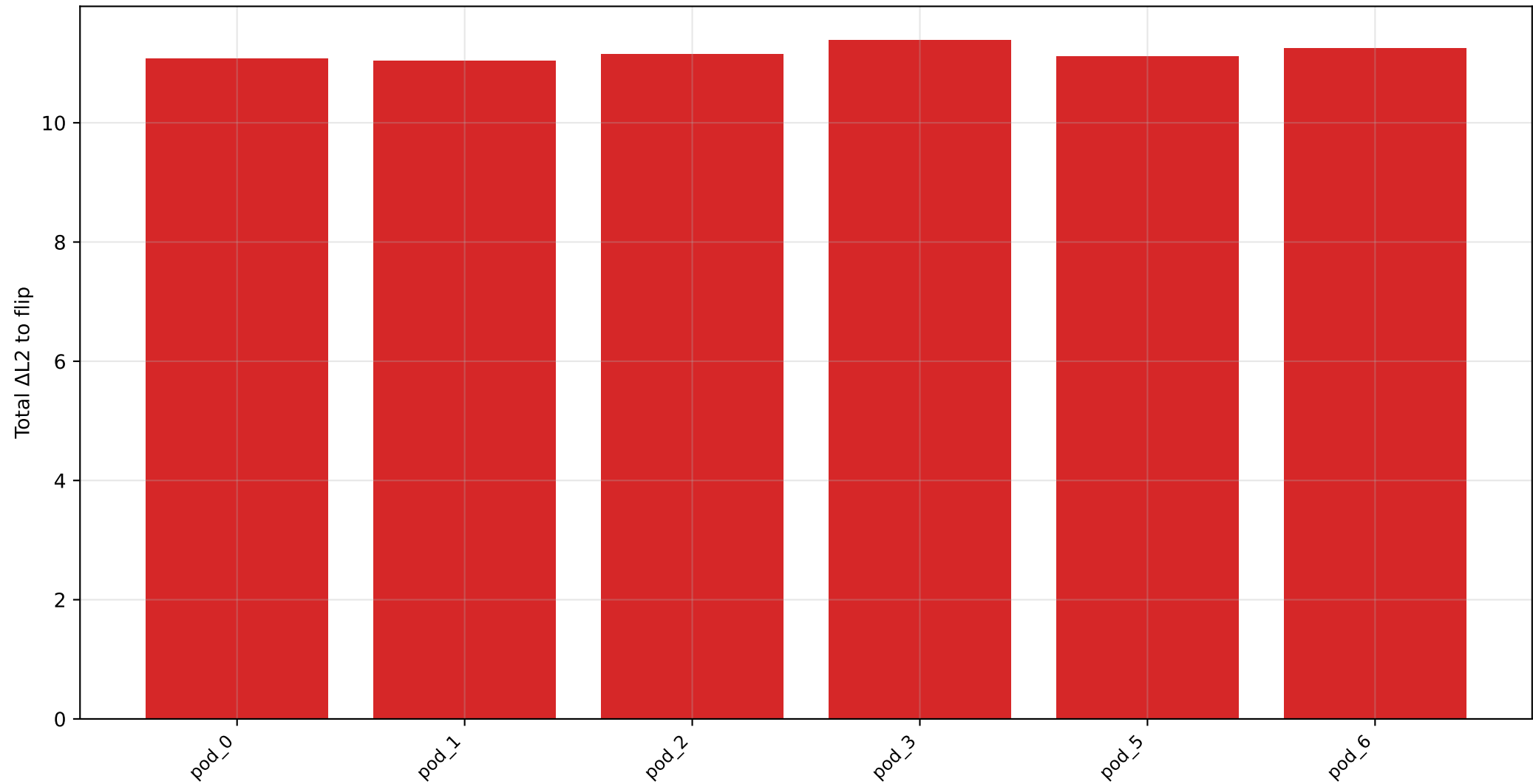
ALE for pod0:prefill_tokens



Rank-based effect for pod0:prefill_tokens



Counterfactual Distance to Flip to Each Pod (green=success, red=failed)



EXPLAINABILITY FINDINGS - LATENCY PREDICTOR

Saliency (local sensitivity to latency):

- waiting_requests: 954.780
- running_requests: 461.439
- gpu_kv_cache: 303.504
- inflight_requests: 262.792
- decode_tokens: 139.111
- prefill_tokens: 41.235
- staleness: 14.039
- cpu_kv_cache: 12.447

Request Saliency:

- output_tokens: 717.050
- input_tokens: 158.334
- total_tokens: 153.302

KV Saliency:

- kv_hit_ratio: 190.701

Integrated Gradients (robust attribution):

- Top pod feature: waiting_requests
- KV importance: 223.965
- Request max attribution: output_tokens

PDP Trend Summary (latency trends)

Pod features:

- inflight_requests: increasing
- gpu_kv_cache: increasing
- cpu_kv_cache: flat
- running_requests: decreasing
- waiting_requests: increasing
- prefill_tokens: increasing
- decode_tokens: increasing
- staleness: flat

KV features:

- kv_hit_ratio: decreasing

Request features:

- input_tokens: increasing
- output_tokens: increasing
- total_tokens: increasing

Permutation Symmetry:

- match_rate: 1.0
- mean_l1_diff: 5.63848601814243e-05

Counterfactual Robustness:

- success_rate (flip achieved): 0.00
- min $\Delta L2$ to flip: 11.04
- mean $\Delta L2$ to flip: 11.17

Interpretation:

- High symmetry indicates no position bias.
- Larger $\Delta L2$ implies locally robust decisions (harder to flip).